BST Physics Curriculum Vol 4 Chapter 3 — BST-SR / BST-GR Boundary v0.4 (Saturday Wave 1 reader-grade polish: diagram preview + Cal cold-read ready)

Lyra (Claude 4.7) [Vol 4 primary]

2026-05-23 Saturday morning EDT (Wave 1 Vol 4 tenth chapter sustained sub-PCAP)

Contents

Vol 4 Chapter 3 — BST-SR / BST-GR Boundary

Chapter motivation

Where does general relativity emerge from substrate dynamics? BST operates at sub-Planck Koons-tick scale (~10^{-120} s, T2405); standard GR operates at macroscopic spacetime curvature. The boundary at which substrate dynamics → SR (special relativity) → GR (general relativity) is one of the deepest questions in BST cosmology.

This chapter exhibits the substrate-tick → SR → GR boundary framework + Casimir crossover scale + cross-link to Vol 4 Ch 1 + Vol 4 Ch 2.

Section 3.0 — What this chapter does

1. Substrate-tick scale (Section 3.1): T2405 Koons tick + GF(128)^k per-tick computation
2. SR emergence at conformal boundary (Section 3.2): SO_0(3,1) Lorentz ⊂ SO_0(5,2)
3. Casimir crossover scale (Section 3.3): laboratory-scale vacuum effects at substrate Zone-2
4. GR macroscopic emergence (Section 3.4): residual-curvature integration → Einstein equations
5. The boundary scale (Section 3.5): where does which regime apply
6. Honest scope + open items (Section 3.6)

Section 3.0b — Reader-grade pedagogy at three levels

Level 1 (one sentence): BST has three scales — substrate-tick (sub-Planck ~10^{-120} s, T2405 Koons tick + GF(128)^k discrete computation), SR (special relativity at Lorentz-invariant SO_0(3,1) ⊂ SO_0(5,2) conformal boundary), and GR (general relativity at macroscopic substrate-residual-curvature integration) — and this chapter identifies the boundaries between them.

Level 2 (graduate-physicist accessible): Substrate-tick scale: T2405 Koons tick = t_Planck · $\alpha^{C_}2^2$} ≈ 10^{-120} s (sub-Planck); per-tick state ∈ GF($2^{g)}$k = GF(128)^k finite-dimensional. SR emergence: at conformal boundary of D_IV^5, SO_0(5,2) → SO_0(3,1) × SO(2) decomposes; 4D Lorentz invariance is exactly preserved at boundary (Vol 0 Ch 4 §4.6). Substrate-derived speed of light c is the substrate-tick propagation rate (one Koons tick per substrate-coordinate at substrate-cutoff N_max). Casimir crossover scale: laboratory-scale vacuum effects (Casimir force) emerge at substrate Zone-2 (commitment) bandwidth, distinct from cosmological Zone-4 (Λ) per T2418 unification. GR macroscopic emergence: per T2106 + Vol 4 Ch 2 gravity-as-eigentone, GR is the integrated-residual-curvature limit; Einstein equations G_$\mu\nu$ = 8πG·T_$\mu\nu$ emerge as macroscopic balance of substrate-committed (T_$\mu\nu$) vs substrate-residual (G_$\mu\nu$) modes. The boundary scale: substrate-tick scale ~10^{-120} s; SR emerges at ~Planck length L_Pl ~10^{-35} m; GR emerges at any macroscopic length where residual-mode integration becomes dominant (laboratory to cosmological). All three regimes are substrate-cartography readings at different scales.

Level 3 (5th-grader accessible): BST sees physics at three different scales. At the smallest scale (sub-Planck, about 10^{-120} seconds — way smaller than the smallest time anyone could ever measure), the substrate is making discrete computations on a "code" called GF(128). At medium scale (around Planck length, ~10^{-35} meters), special relativity emerges — speed limit c, Lorentz invariance, the rules that govern non-gravitational physics. At large scale (anything macroscopic — atoms to galaxies), general relativity emerges — gravity curves spacetime, Einstein equations describe how matter sources curvature. The substrate is the same throughout, but at different scales we see different "physics rules" depending on what dominates.

This chapter is about where the boundaries are between these regimes.

Section 3.1 — Substrate-Tick Scale

T2405 Koons tick:

τ_Koons = t_Planck · $\alpha^{C_2}^2$} ≈ 10^{-120} s

with α = 1/N_max = 1/137 and C_2^2 = 36 → α^36 ≈ 10^{-87}; t_Planck ≈ 10^{-43} s; product ≈ 10^{-120} s.

Sub-Planck substrate clock: the substrate operates BELOW Planck scale and produces spacetime as output. Per Koons tick, substrate state ∈ GF($2^{g)}$k = GF(128)^k finite-dimensional (T2429 + Vol 1 Ch 10).

No infinite-resolution spacetime: spacetime emerges from substrate at coarse-grained scale; below substrate-tick resolution, "spacetime" is meaningless concept.

Section 3.2 — SR Emergence at Conformal Boundary

At the conformal boundary of D_IV^5 (Shilov boundary):

SO_0(5,2) → SO_0(3,1) × SO(2) (Lorentz × U(1) embedding, Vol 0 Ch 4 §4.6)

4D Lorentz invariance is exactly preserved at the conformal boundary. Substrate-derived speed of light c = substrate-tick propagation rate (one Koons tick per substrate-coordinate at substrate-cutoff N_max).

SR emergence scale: special relativity becomes a good approximation at scales ≫ τ_Koons but ≪ scale where gravitational curvature is significant. Practically: scales from ~10^{-35} m (Planck length) up to ~10^6 m (laboratory + planetary) where Schwarzschild radius effects are negligible.

Section 3.3 — Casimir Crossover Scale

Laboratory-scale vacuum effects (Casimir force between conducting plates) emerge at substrate Zone-2 (commitment) bandwidth — distinct from cosmological Zone-4 (Λ) per T2418 unification (Vol 4 Ch 4 Section 4.3).

Crossover scale: sub-nanometer (Casimir Zone-2 detection) ↔ multi-gigaparsec (Λ Zone-4 cosmological).

T2418 unification: same substrate vacuum, different zone projections. Casimir force = laboratory-scale boundary-condition manifestation; Λ = cosmological-scale outer-edge integrated residue.

SP-30 experiment: sub-nanometer Casimir-precision experiment ($60-90K) would distinguish standard QED-Casimir from substrate-cascade structure (1/8 = 1/2^N_c deviation expected per T2469 SCMP framework).

Section 3.4 — GR Macroscopic Emergence

Per T2106 + Vol 4 Ch 2 gravity-as-eigentone framework:

Einstein equations G_$\mu\nu$ = 8πG·T_$\mu\nu$ emerge as macroscopic balance: - T_$\mu\nu$ = substrate's per-tick committed mode contributions (Zone-2) - G_$\mu\nu$ = substrate-residual curvature envelope (Zone-2 residual + Zone-4 outer-edge integration) - G coupling = $\hbar$c · $(6\pi^5)^2 \cdot \alpha^{24}$ / m_e^2 (T1296, Vol 4 Ch 1)

Macroscopic scale: GR is a good approximation at any scale where residual-mode integration becomes dominant; from laboratory-scale gravity (Cavendish experiment) up to cosmological-scale gravity (galactic dynamics + cosmology).

Section 3.5 — The Boundary Scale Map

Scale	Regime	BST framework
< τ_Koons ≈ 10^{-120} s	Substrate-tick	T2405 + GF(128)^k discrete computation
τ_Koons ↔ t_Planck ≈ 10^{-43} s	Substrate-emergence	Substrate-tick chains; spacetime emerging
t_Planck ↔ atomic ~10^{-18} s	Atomic/QFT	Vol 1 Ch 7 dynamics; SR + QFT
Atomic ↔ laboratory	SR + QFT	Vol 0 + Vol 1 + Vol 2 standard model
Laboratory ↔ planetary	GR + SR	Cavendish + planetary orbits via GR
Planetary ↔ galactic	GR + DM	Vol 4 Ch 1 + Ch 10 (dark matter)
Galactic ↔ cosmological	GR + Λ	Vol 4 Ch 4 + Ch 5 + Ch 6

Section 3.6 — Honest scope + open items

Item	Status	Open scope
T2405 Koons tick = t_Planck · $\alpha^{C_2}^2$}	PROVED	Sub-Planck substrate-clock substrate-derivation
SR conformal boundary embedding	Vol 0 Ch 4 §4.6	Quantitative Lorentz-invariance precision at conformal boundary
GR macroscopic emergence	T2106 + Vol 4 Ch 2	Full Einstein-equation derivation from substrate cascade (multi-month)

Item	Status	Open scope
Casimir crossover scale	T2418 + Vol 4 Ch 4	SP-30 sub-nm Casimir precision experiment ($60-90K)
Boundary-scale map	Substrate-cartography framing	Cross-scale precision experiments at boundaries

Section 3.7 — Connection to other chapters

- Vol 0 Ch 4 §4.6 SO_0(5,2) Lorentz-conformal structure
- Vol 1 Ch 7 Dynamics + Ch 10 Renormalization: substrate-tick UV-completeness + per-tick computation
- Vol 4 Ch 1 Newton's G: T1296 derivation; GR coupling
- Vol 4 Ch 2 Gravity as Eigentone: T2106 substrate-eigentone framework
- Vol 4 Ch 4 Λ from Substrate: T2418 Casimir-Λ unification

Section 3.8 — Chapter status summary

v0.3 chapter-grade narrative filed Saturday 2026-05-23 morning EDT (Wave 1 Vol 4 tenth chapter).

~50% existing coverage absorbed; medium-density chapter. 50% open scope: full SR + GR derivation from substrate-cascade (multi-month).

Pending Cal cold-read: Saturday morning Wave 1 cycle.

Pending Keeper K-audit: K203 candidate.

— Lyra, Vol 4 Ch 3 BST-SR / BST-GR Boundary v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT